

Kubernetes Architecture

What is Kubernetes Architecture?

Simple Definition

Kubernetes Architecture is the structure of Kubernetes that shows how all its components work together to manage containers.

Beginner Definition

Kubernetes Architecture is the way Kubernetes is organized to deploy and manage applications automatically.

Easy Real-Life Example 🏠

Imagine a school.

- 🏠 Principal = Kubernetes Control Plane (Master)
- 👩‍🏫 Teachers = Worker Nodes
- 🎓 Students = Pods (Applications)

The principal **doesn't teach students**.

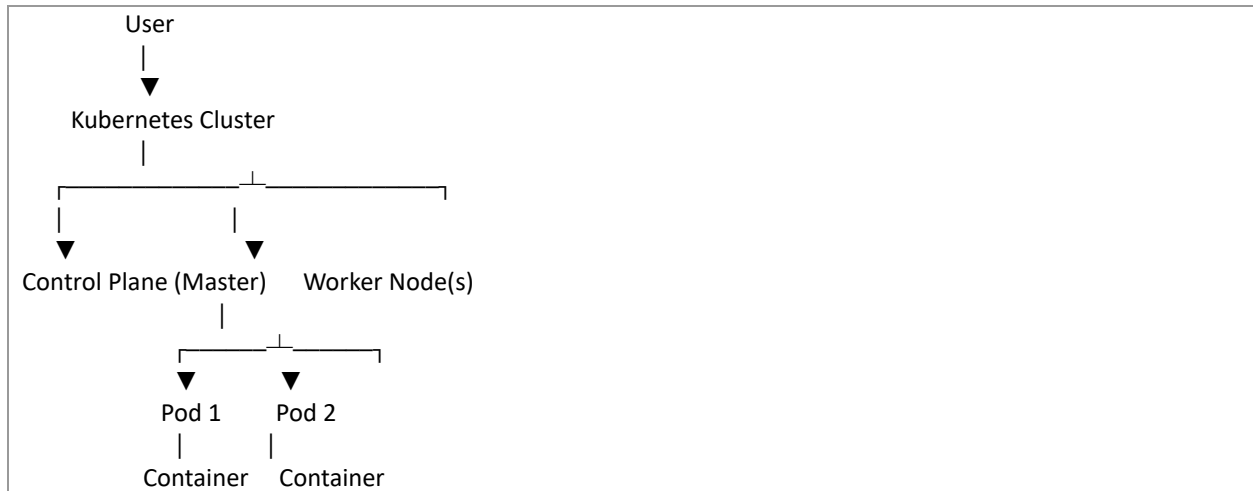
Instead, the principal:

- Gives instructions
- Assigns teachers
- Checks attendance
- Solves problems

Teachers do the actual teaching.

Kubernetes works in exactly the same way.

Basic Kubernetes Architecture



Kubernetes Architecture (According to the Figure)

First look at the figure

Kubernetes Cluster

M W1 W2 W3

Where:

- **M = Master Node (Control Plane)**
- **W = Worker Node**
- All together = **Cluster**

Step 1: What is a Cluster?

Definition

A Kubernetes Cluster is a group of computers (nodes) working together to run containerized applications.

Easy Meaning

Instead of using one computer,

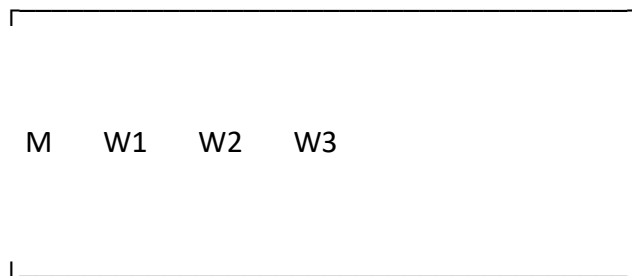
Kubernetes connects many computers.

These computers are called **Nodes**.

Together they form a **Cluster**.

According to the Figure

Cluster



Here,

- 1 Master Node
- 3 Worker Nodes

Together they make one Kubernetes Cluster.

Step 2: Master Node (M)

Definition

The Master Node (Control Plane) manages the entire Kubernetes Cluster.

Easy Meaning

Think of the Master Node as the **Manager** or **Brain**.

It **does not run applications**.

It only manages everything.

Responsibilities

The Master Node:

- Receives user requests
 - Decides where Pods should run
 - Monitors Worker Nodes
 - Creates new Pods
 - Restarts failed Pods
 - Manages the whole cluster
-

Real-Life Example

Imagine a company.

Manager



Employees

The manager:

- Assigns work
- Checks employees
- Solves problems

Employees actually do the work.

Similarly,

Master Node manages everything.

Step 3: Worker Nodes (W)

Definition

Worker Nodes are machines that actually run applications inside Pods.

Easy Meaning

Worker Nodes are employees.

They receive work from the Master Node.

Responsibilities

Worker Node

- Runs Pods
 - Runs Containers
 - Runs Applications
 - Sends status back to Master
-

According to the Figure

M

↓

W1

↓

Application Running

M

↓

W2



Application Running

M



W3



Application Running

What is inside Worker Node?

Every Worker Node contains

Worker Node



Pods



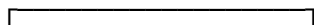
Containers



Application

Example

Worker Node



Pod



Container



Nginx App



Complete Flow

User



Master Node



Worker Node



Pod



Container



Application

Who Manages Orchestration?

This is a very common KCNA interview question.

Answer

The Kubernetes Control Plane (Master Node) manages container orchestration.

What is Orchestration?

Simple Definition

Orchestration means automatically managing containers.

The Master Node performs orchestration.

How does the Master Node manage Orchestration?

Suppose

100 users visit your website.

The Master Node notices that traffic has increased.

↓

It automatically creates more Pods.

This is orchestration.

Suppose

One Pod crashes.

Master Node notices.

↓

Creates another Pod automatically.

Again,

This is orchestration.

Suppose

Worker Node 2 becomes full.

Master Node says

"Don't place any more Pods here."

↓

Places the next Pod on Worker Node 3.

Again,

Orchestration.

What things does Kubernetes Orchestrate?

The Master Node automatically manages:

✓ Deploying applications



✓ Creating Pods



✓ Scheduling Pods



✓ Monitoring Pods



✓ Restarting failed Pods



✓ Scaling Pods



✓ Load Balancing



✓ Rolling Updates



✓ Rollback



✓ Networking



✓ Storage

All of this together is called **Container Orchestration**.

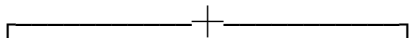
Easy Diagram

USER



MASTER NODE (Control Plane)

(Brain / Manager)



Worker 1 Worker 2 Worker 3



Pod

Pod

Pod



Container Container Container

| | |

Application Application Application

Easy Real-Life Example

Imagine a restaurant.

 Restaurant Manager = Master Node

 Chefs = Worker Nodes

 Food = Application

Customers place an order.

↓

Manager decides

"Chef 1 will cook this."

↓

Chef 1 cooks the food.

If Chef 1 becomes busy,

↓

Manager sends the next order to Chef 2.

If Chef 2 leaves,

↓

Manager immediately assigns another chef.

This automatic management is called **Orchestration**.

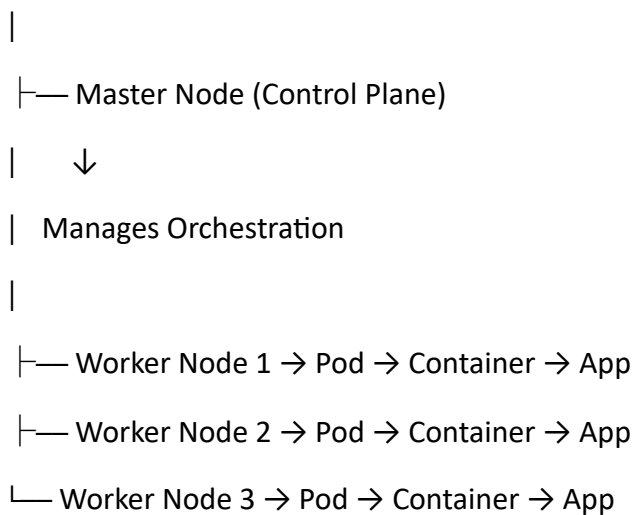
Quick Revision

Component	Easy Meaning
Cluster	A group of Master and Worker Nodes working together
Master Node (Control Plane)	The brain or manager of the cluster
Worker Node	Runs Pods and applications
Pod	The smallest unit Kubernetes deploys
Container	Holds the application
Container Orchestration	Automatically managing containers and Pods

Who manages orchestration? The Kubernetes Control Plane (Master Node)

★ Remember This Formula

Cluster



Easy Memory Trick:

- 🧠 **Master Node = Brain = Makes Decisions**
- 💻 **Worker Node = Does the Work**
- 📦 **Pod = Runs the Container**
- 📦 **Container = Runs the Application**
- ⚙️ **Kubernetes = Orchestrates (Manages) Everything Automatically**

